Ground shakes: cause still unresolved

ProtoDUNE-HD · Manuel Arroyave · Analysis through 23 October 2026

We checked seven runs.

2024 data	Run	Events
July	27938	6
	27951	19
October	30201	44
	30202	83
November	32730	1
	32731	716
	32739	662
Total		1,531

Selected saved records. Automatic candidates are not confirmed shakes or a full-run rate.

We used charge and PDS data.

TPC: charge readout. PDS: photon detector readout.

We also checked July voltage records.

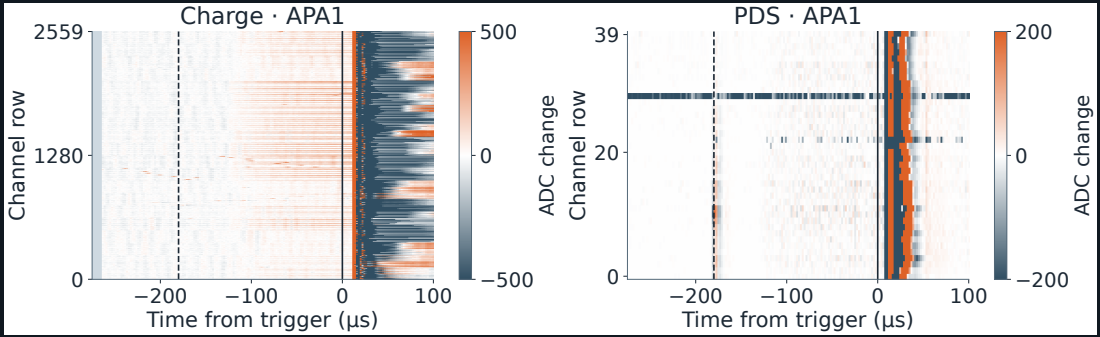
Charge test: 1,516 possible shakes.

Five records did not pass the test; ten lacked usable charge data.

- Charge signals shift in opposite directions.
- Disconnected PDS inputs also respond.
- No unique charge–light match is established.

A PDS pulse comes before the large disturbance

Run 27938 · event 20576 · APA1 charge and continuous PDS readout

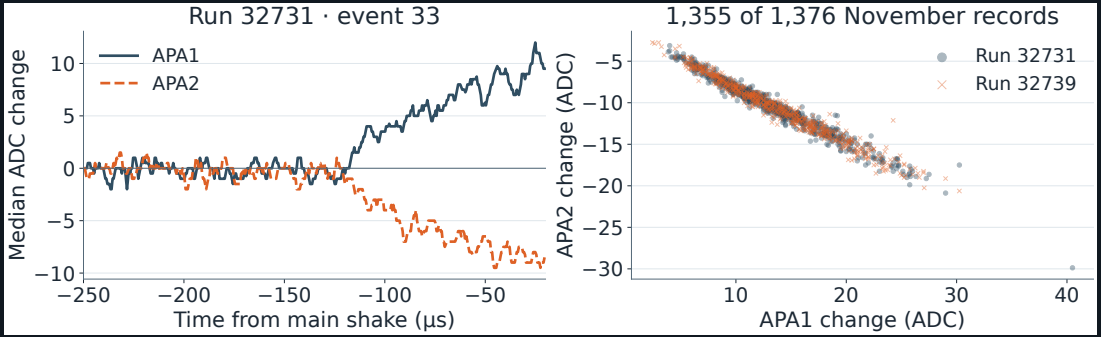


Each row is one channel. Blue is below baseline; orange is above it. The dashed line marks an early PDS pulse, about 180 μs before the trigger (0). Both systems then show a large disturbance. Grey means unrecorded.

ADC counts are raw signal units; μs means millionths of a second. Colour limits clip large signals. Recorded timing; electronics delays uncorrected.

The slow charge signals move in opposite directions

APA1 and APA2 are the two charge-readout assemblies used here

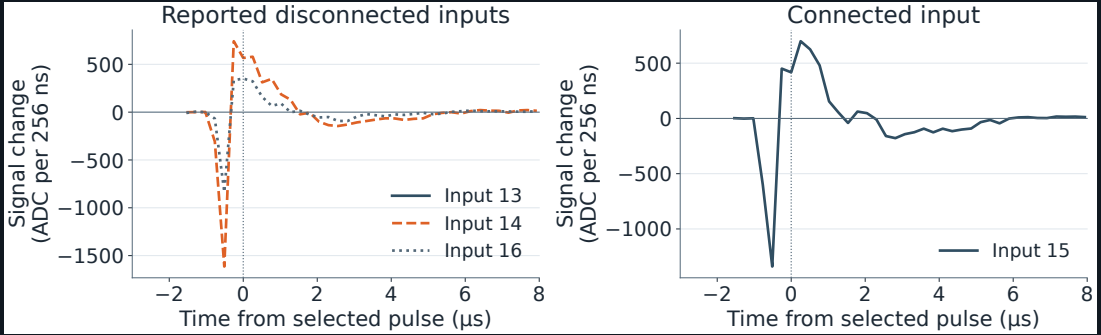


Left: one event. APA1 rises while APA2 falls before the main shake. **Right:** one dot per event. This pattern is clear in 1,355 of 1,376 November records.

The slow signal must exceed the noise cut in both assemblies. Zero time is a charge-threshold reference, not the first physical disturbance.

Disconnected PDS inputs also carry pulses

Run 32731 · event 33 · selected pulse 183.296 μs before the main shake

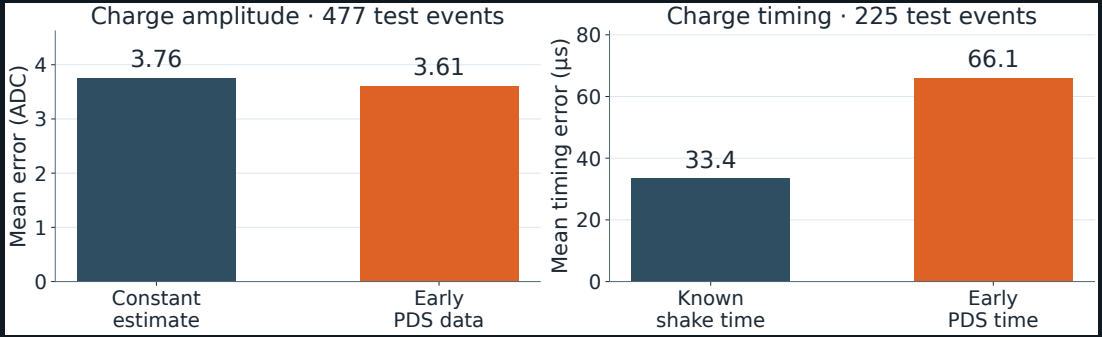


Left: inputs 13, 14 and 16 were reported disconnected from the sensors. **Right:** connected input 15 responds at the same time. This supports electrical pickup. It does not identify what started the pulse.

Each line shows signal change in 256 ns steps. The reported disconnections still need run-specific confirmation.

Early PDS signals give weak predictions

Trained on run 32731; tested on unused run 32739 events · Lower bars are better

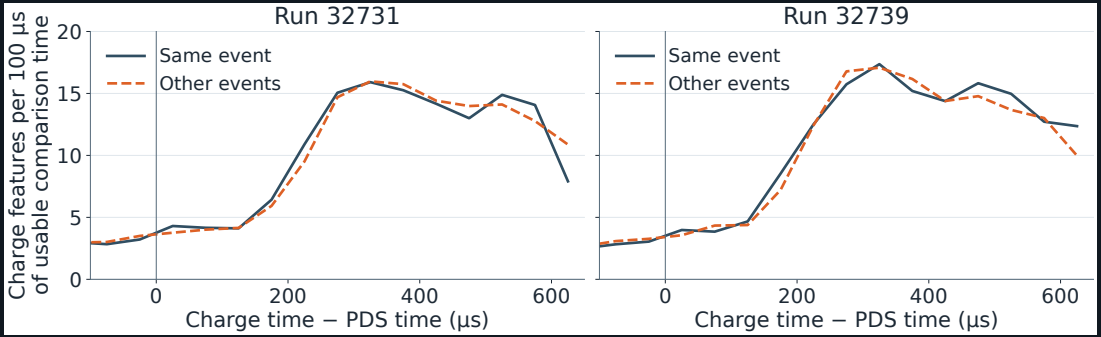


Left: early PDS data reduce the charge-amplitude error by only 4.1%. **Right:** the early pulse predicts the 2 ADC charge crossing less accurately than the known main-shake time. These checks use saved events.

This tests variation within selected shakes. It does not test whether PDS can predict that a shake will occur.

We found no unique charge–light match

Early PDS pulses compared with later charge features · November runs

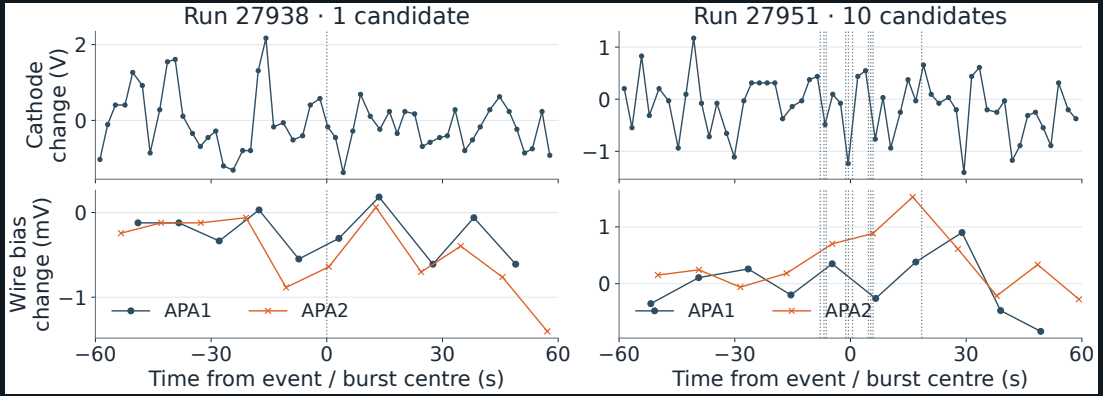


The horizontal axis is charge time minus PDS time. Blue uses pulses from the same event; dashed orange uses pulses from other events. Both show the peak near 325 μs. That peak alone does not link a PDS pulse to a charge track.

Counts account for recorded comparison time. Pulses are swapped within the same run, file and assembly. This is not a normal-running background test.

July voltage records show no large ramp or trip

Eleven July candidates; ten occur in one 26-second burst



Top: cathode voltage change. **Bottom:** wire-bias change. Dots are archived samples; vertical marks are shake candidates. Sampling every 2–11 seconds cannot rule out a brief discharge.

Changes are relative to the preceding baseline: about 176 kV for the cathode and -370 V for the wire bias.